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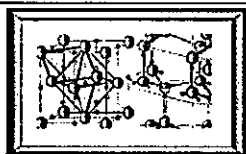
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COLD FUSION TIMES

CETI's MICROSPHERES AND REACTOR "DEMONSTRATED"

Anaheim, California - The December '95 POWER-GEN conference may have been the world's largest electric power industry trade show. Over 850 exhibitors and 14,000 people from 75 countries attended the meeting. The array of 250 conference paper presentations included topics such as gas turbines, electric systems, government regulations, control systems, and fluidized beds.

Although controversial, and discussed later in this edition, some cold fusion enthusiasts have claimed that the cold fusion technology available today will make competing energy sources "obsolete as the buggy whip and the slide rule". Will that technology include *CETI's* microspheres? perhaps some other technology? This issue of the **COLD FUSION TIMES** presents several parallax views of the same *CETI* demonstration as well as critical follow-up technology and business reports. Several points of view are presented. Analyses of the thermometry, calorimetry, and the material science are reported beginning on pages 3, 6 and 7.

DISAPPOINTED - UNDAUNTED - COLD FUSIONEERS MEET

Cambridge, MA - The 1996 MIT annual IAP Conference on Cold Fusion met this year at the Marriott Hotel in Cambridge. Although several MIT Alumni previously worked to make this annual meeting available without charge at MIT IAP, this time some of them finally avoided the annual hassle and shifted the formal setting far away from the Institute's "Infinite Corridor". The meeting was launched by Dr. Eugene Mallove and Jed Rothwell and included discussions about *CETI*. The audience clearly had hoped to examine a demonstration setup. However, previous *CETI* proponents were left to flail tentative possible reasons why *CETI* had suddenly clamed up from its previous open pattern of public demonstrations revealing some of their microsphere technology.

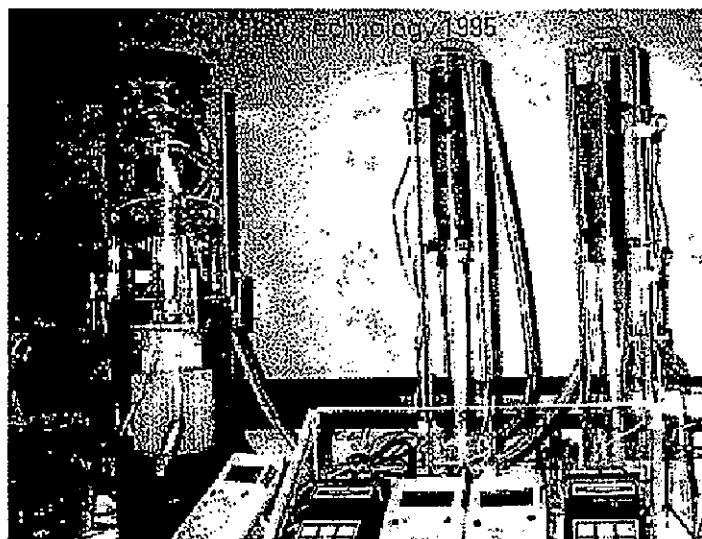
The Saturday meeting of scientists (median age appeared to be 50 years) numbered over one hundred. They were attentive during the entire day as the forum's focus shifted from the *CETI* brouhaha toward material and nuclear science and then to

(continued page 2)

CETI's RELATIVE SILENCE

Dallas, Texas - Our call to Jim Reding at *CETI* was made to give him an opportunity to clarify the various points of view. He stated that there is not a company write-up or fax available, but that "people would be provided updated information" toward the end of March 1996. He stated that his company relies upon Jed Rothwell, "who took the measurements" during most of the public series of experiments, and that other laboratories were now conducting additional tests.

Mr. Reding, President of the company, stated that they are producing 1000 Watts out (for a flow rate of about a liter per minute) for 1 watt in. The observed delta-T, he said, was in the range of 15 to 20% and was apparently a function of flow rate. He continued that his position would be one of silence at this time because of both corporate interests and because *CETI* is developing "a product".



CETI/Patterson Setup at Anaheim

The cooling tower is on the left hand side with the reactors reported to be on the right side - the CF cell and control cell respectively. The control box is barely seen on the lower left. The differential thermocouples and ammeters are at the bottom. [see page 2 (credits), and pages 3, 7]

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TOWARD THE FUTURE

1996 CURRENT STATE OF COLD FUSION

by Edmund Storms

This brief excerpt is from a November '95 MEMORANDUM to Tom Claytor from Dr. Storms which was picked up through the Internet. It is important here because it is yet another view of the *CETI* demonstrations. The excerpt is also important because the **COLD FUSION TIMES** supports Dr. Storms' solid reporting of scientific breakthroughs, and because it summarizes significant events in 1995, which supplement the material and information previously presented in the **COLD FUSION TIMES**.

"Several interesting events have occurred since we talked last. They are summarized below with details following.

1. The Pons-Fleischmann patent has been issued by the European Union.
2. The Piantelli patent has been issued by Italy.
3. *MITI* (Japan) has increased its funding for Cold Fusion (New Hydrogen Energy) from \$30M/4 years to \$100M/4 years because of significant success.
4. The *CETI* (Patterson) light-water device has been tested by 3 major laboratories and found to produce excess energy. A factor of 10 excess is routine with levels of 70 times being recently reported.
5. The American Nuclear Society arranged for a panel discussion on Cold Fusion at the recent SOFE meeting.
6. The *Fiat Auto Company* arranged for an all expense paid meeting of experts on Cold Fusion at Asti (Oct. 11-13, 1995) in Italy which I attended.
7. My self-funded efforts to make excess energy using palladium supplied by *IMRA*(Japan) have succeeded in producing 10% excess (4 W max).
8. The Third Conference on Cold Fusion and Nuclear Transmutation (RCCF-3) took place in Dagomis (Russian Federation) on Oct 2-7, 1995.

The Patterson light-water cell was demonstrated at the SOFE meeting after being assembled and tested in Prof. George Milev's laboratory at the University of Illinois. They were able to generate 4 W, of excess power by applying 0.4 W. The recombination energy is

not included. This device has obtained the attention of a major company where it was tested with success. This company is now going to acquire *CETI*. The results of the deal will be announced in the next few months.

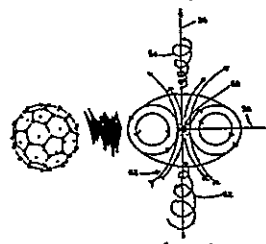
A panel discussion on CF was held at the SOFE meeting (Oct. 1995). Much interest was shown with only a few people revealing extreme skepticism. With the hot fusion budget dropping, people are looking for any port in the storm. Unlike the attitude in US government and at LANL, interest in learning about the phenomenon is growing elsewhere." >

FUSIONEERS MEET

(continued from page 1) business and other over-unity devices under examination by experimentalists. The meeting featured Hal Fox, Keith Johnson, and James Griggs each outlining advances in their technologies. Dr. Johnson showed a preview of his movie Excess Heat and shared more details of the plotline. Mr. Griggs shared further information including the fact that NASA will now investigate his device.

Even though absent, much talk was about *CETI* and its retreat since the POWER-GEN conference until even over the Internet comments were posted regarding the unavailability of their "demonstration". Jed Rothwell seemed the most disappointed. He had nurtured the company's PR, after he serendipitously "discovered" the company's existence through a Japanese newspaper! Mr. Rothwell had reported on two groups of demonstrations and felt that *CETI* had a "business relations fiasco". He had good ideas regarding future instrumentation for them, and was shocked by their "interruptus school of public relations". One rumor circulated is that *Motrola* was one of the companies that may have produced the sudden change in public relations behavior. >

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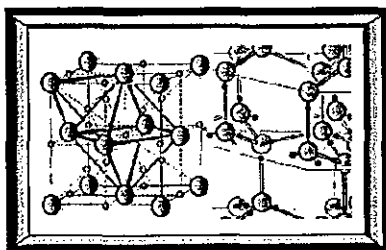
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CETI Setup at Anaheim - Photos (page 1 and 7) from John Logajan's Web site, taken by Akira Kawasaki & Jed Rothwell.
Back Cover: Gaseous Pillars M16. Taken with HST-WFPC2 PRC95-44a J. Hester, P. Scowen (NASA, 11/2/95)
Cold Fusion Times LOGO - Palladium must be fully filled with deuterons to generate the reactions. The logo was in part based upon a water structure drawn by John Mara. In the logo, the face-centered cubic fully loaded metallic structure is shown on the left, totally filled with deuterons obtained from the heavy water on the right hand side.
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PEOPLE IN THE NEWS Winter '96

RETURNS NEXT ISSUE - We bring you clear excerpts of opinion this month about CETI

MORE THOUGHTS ON ANAHEIM DEMO

ANAHEIM VIEWED BY CETI's CHIEF SUPPORTER

by Jed Rothwell

Atlanta, December 12, 1995 - Last week at the Power-Gen '95 Americas power industry trade show in Anaheim (December 5-7, 1995), a 1-kilowatt cold fusion reactor was demonstrated by *Clean Energy Technologies, Inc. (CETI)* of Dallas Texas. The cathode is composed of thousands of 1 mm diameter co-polymer beads with a flash coat of copper and multiple layers of electrolytically deposited thin film nickel and palladium. CETI holds four U.S. patents on the beads, with additional patents pending. During the demonstration, between 0.1 and 1.5 watts of electricity was input, and the cell output 450 to 1,300 watts of heat.

In April 1995, at the Fifth International Conference on Cold Fusion (ICCF5) CETI demonstrated a cell with input of 0.14 watts and a peak excess of 2.5 watts, a ratio of 1:18. In October 1995, at the 16th biannual Symposium on Fusion Engineering (SOFE '95) the University of Illinois showed a CETI cell with 0.06 watts input and 5 watts peak output, a ratio of 1:83. Ratios at Power-Gen ranged from 1:1000 to 1:4000.

The ICCF5 and Power-Gen calorimeters were designed and constructed by Dennis Cravens. The SOFE '95 calorimeter was constructed by George Miley's group at the University of Illinois. The Power-Gen cell and calorimeter are much larger than CETI's previous cold fusion demonstration devices. The cell is 10 cm long, 2.5 cm in diameter, containing roughly 40 ml of beads. Previous cells had about 1 ml of beads. The cell itself is wrapped in opaque foam plastic because the cell geometry has been improved and the improvements are not yet covered by patent applications. Other components in the calorimeter are made of clear Lucite plastic.

The flow calorimeter reservoir holds 2.5 liters and the flow rate is set between 1.0 and 1.5 liters per minute. A control cell is mounted parallel to the hot cell. The flow to both cells is regulated with precision valves. The reservoir and pump consist of a Magnum 220 aquarium pump with a micron filter attachment, with an additional Lucite cylinder built on top of the pump unit to hold a cooling coil, gas trap, and a 3.5 watt computer cooling fan. Water is circulated by a magnetic impeller pump,

(continued page 4, see also page 7)

DISTANCE BETWEEN ROTHWELL AND CETI

by Richard A Blue

I commend Mitchell Jones for his demonstration (Ed. see page 6) that there may well be problems with the CETI demonstration with regard to the heat dissipation capabilities of the "radiator." What is most clearly being brought out in subsequent discussions of this matter is that the information that has been provided is not really adequate to reach any sound conclusions. Even with regard to the supposed operating power level of the CETI device the numbers are all over the place. At first we were fed a claim that "1300 watts" was clearly demonstrated as if that level had been sustained and/or replicated for a significant number of hours. Now that appears to be contrary to fact. The 1300 watt operation was some sort of maximum, and we can't even get a straight answer as to how long that maximum was observed. If, as the Jones test indicates, the cooling capacity of the radiator was likely inadequate for 1300 watts there should have been a clear signature in the experimental data.

In fact, the performance of the radiator should have had an effect on the data even if every thing worked perfect or could be judged if just one measurement had been logged! All we need to know is the temperature of the fluid leaving the radiator. Amazingly this temperature reading has been ignored as not having any particular significance toward an understanding of the CETI demonstration. Simply stated, the extraction of more heat from the fluid in the radiator must imply a rise in the temperature of the fluid exiting the radiator. It is flow calorimetry in reverse. When the temperature differential across the Patterson cell goes up there must also be an increase in the temperature differential across the radiator--unless there is something bogus about the temperature differentials in the first place. >

Here is some sample data as reported by Mr. Rothwell, of the Patterson/Cell:

Test 1, December 4, two hours

INPUT POWER

Measured AC: 0.7 A * 120 V = 84 W

Electrolysis: 0.18 A * 8 V = 1.4 W

OUTPUT POWER

Flow rate 1200 ml/minute (300 ml in 15 seconds)

Delta T Temperature 16 to 17 deg C

1200 ml * 16 deg C * 4.2 = 80,640 j/min = 1,344 W

Test 2, December 5, afternoon, 30 minutes.

INPUT POWER

Measured AC: 0.7 A * 140 V = 98 W

Electrolysis: 0.02 A * 3.9 V = 0.1 W

OUTPUT POWER

Flow rate 1000 ml/min (250 ml/15 seconds)

Delta T Temperature 6.7 deg C

1000 ml * 6.7 * 4.2 = 28,140 j/min = 469 W

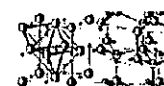
OF GREAT INTEREST
TO OUR READERS
SEE FOR THE WORLD (PAGE 8)

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CETI's 1.3 KILOWATT REACTOR DEMONSTRATED!

by Robert Bass

Anaheim, California - December 4, 1995 will be remembered in history like Fermi's epochal first fission pile at the U of Chicago. On this date a group of unaffiliated scientists witnessed a demonstration of the 1.3 kW steady-state 'CF' reactor produced by CETI Inc. under the Patterson Power CellTM patents, which included a simultaneously-operated control demonstration that should silence all critical objections, and by any reasonable scientific criterion produces 960 times more thermal energy than electric energy consumed. Specifically the 'active' electrolytic cell used a 1.4 Watt input (namely 0.18 Amps at 8 Volts) and caused a full 16 degrees Centigrade rise in the temperature of water flowing at the rate of 1.2 liters/sec (i.e. 1,344 Watts).



Present at this private preview of the public demonstration to be made before the 14,000 attendees at Power-Gen '95 on December 5 were, of course, Dr. James Patterson, inventor, and his grandson Jim Reding, President of CETI Inc. The setup was explained and questions fielded by Dr. Dennis Cravens, who answered every question asked except one pertaining to a technical improvement (now patent pending) in the internal details in comparison to the demonstration made at SOFE (which has been independently replicated by graduate students of Dr. George Miley, Editor-in-Chief of the American Nuclear Society's journal 'Fusion Technology'). Except for this one novel improvement, all details were similar to those disclosed in the 4 issued patents and previous news releases. But this new wrinkle has improved the 'gain' from 80-to-1 to 960-to-1!

Dennis Cravens admitted that 'CF' is a generic term, and that if e.g. Hal Puthoff is right about ZPF energy, the excess energy need not be nuclear. My personal speculation is that the nickel lattice is catalyzing the fusion of two protons into a deuteron plus a positron [soon annihilated by collision with an electron, making a photon]; when I asked Jim Reding "Supposing that the reaction is nuclear, how long would you have to run this cell before the build-up of heavy water would become easily measurable?" he replied: "We are leaving that calculation to people like you to figure out." At one point Dennis wrote on the board symbols indicating that if one started with a spin-aligned proton-plus-Lithium-7 pair (i.e. used magnetically-polarized fuel), then there is an enhanced probability of fusion to yield an excited Beryllium nucleus, which spontaneously decays into two alpha-particles (helium-4 nuclei), but it was not clear from his amused expression whether he was giving us slow learners a hint, or simply tweaking the competition with deliberate disinformation. (He has previously published articles about spin-polarization enhancement in 'FUSION FACTS', mentioning that this is a known concept in hot fusion circles, so I surmise he was serious.) On seeing

the setup operating with my own eyes, and excluding the far-fetched possibility of deliberate fraud, i.e. accepting the readings from the multimeters and other instruments as valid, the hypothesis of experimental error or accidental self-deception seems to me absurdly remote.

The much-discussed pump was circulating water through two parallel vertical columns; we twice measured that 300 milliliters of water was being pumped through each columnar cell in 15 seconds, making 1.2 liters/minute. The control cell was drawing 1.6 Watts (specifically, 0.33 Amps at 4.75 Volts) and contained what was called '10-shot' metallic beads. The rise in temperature of the water through the control cell was a paltry 2.6 degrees Centigrade. CETI would not comment on any improvements to the cell design or the beads contained inside the cell. All that was said was that the active cell contained "less than 40 milligrams of metal" which was not specified. As mentioned above, the active cell was drawing 1.4 Watts (namely 0.18 Amps at 8 Volts), and so [using the standard specific heat conversion factor of 69.8] if one multiplies 16 degrees Centigrade by 1.2 liters/minute one gets 1.344 kilowatts of thermal output. Dividing 1,344 by 1.4 yields a 'gain' of 960!

Skeptics who want to whine that power is sneaking from one end of the table, where the pump motor is running, all the way to the active column and somehow influencing the active cell but not the immediately adjacent parallel control cell should explain the miracle of 'selective sneaking'. However, to give the Devil's Advocate his due, the witnesses noted that the total power involved in the entire experiment, pump and both cells and all, was only 84 Watts, so the skeptics' gain may be defined as 1,344 divided by 84, namely 16-to-1, which is still a remarkable technological breakthrough any way you want to think about it! I believe that this preview-demonstration was one of the high points of my 65 years on planet Earth -- as memorable as when I met Einstein at a 1950 Rhodes Scholars' sailing party! ➤

Rothwell on CETI

(continued from page 3)

driven by a 50-watt motor mounted underneath. Static in-line mixers ensure mixing. (These are plastic objects about an inch long with vanes to stir the flow.) A few weeks before the conference, Cravens decided to increase the flow rate in order to keep the temperature below 50 degrees C. The new flow rates exceeded the capacity of his flowmeters. He was not able to procure a bigger flowmeter in time for the conference, so no flowmeter was installed. Flow was measured by turning stopcocks to redirect fluid from the cell outlet tube into a graduated cylinder for 15 seconds. This test was performed many times, and the flow rate was not observed to change measurably, except when it was deliberately adjusted between runs. The water hose from the pump is coiled in an air cooled box on top of the reservoir. Air is drawn through the box by the cooling fan. The pump, cooling fan and DC power supplies electrolysis all have one common AC cord, which is monitored by a Radio Shack analog AC voltmeter and a multimeter. Total power consumption by all components is 85 watts. (continued page 5)

MATERIALS CORNER

HELIUM-4 (He^4) PRODUCTION DURING COLD FUSION REACTIONS

(Part III)

by Mitchell Swartz

Cold fusion reactions are quite novel because they only occur in a physically dense material and not in a "gas-like" hot fusion plasma. They are also novel because they create helium-4, which is one of the major products (or nuclear "ashes") of this new technology. No known chemical reaction is known to generate de novo quantities of helium. The only sources of more helium are those which can be created with either nuclear fusion in hot systems or by the radioactive decay of tritium (which occurs by emission of an electron), and apparently -- to the consternation of the scientific world -- cold fusion. Unlike hot fusion which generates helium-3 (and neutrons), cold fusion generates harmless helium-4, an inert gas which is already in the atmosphere, and is even used to filled party balloons.

Cold Fusion's product : helium-4

Helium is a quite minor component of the Earth's atmosphere. The amount of helium in air ranges from 6 to 33 parts per million. However, in some natural hydrocarbon gases helium is contained up to a relatively whopping 6-7%. There are two types (isotopes) of helium, and both are peculiar in most of their physical and chemical properties. Helium is an inert gas because it does not form compounds with other elements. In the Earth's atmosphere, the majority of the helium (99.99987 %) is helium-4 (He^4) which is composed of 4 atomic units (two protons and two neutrons). The rest is a very minor fraction called helium-3 (He^3) which contains the two protons required for helium but, unlike helium-4, only one neutron.

As previously discussed (Parts 1,2), some of the remarkable observations in the field of cold fusion clearly demonstrate that the helium-4 "ash" production by cold fusion is consistent with some of the reactions having a probable nuclear origin. What is most compelling is that the helium-4 production is time-correlated with the onset of excess heat, to its observed power levels of excess heat, and even correlated with the exposure of undeveloped x-ray films which are blanketed around some active electrodes. Furthermore, the use of ultrasound can also create helium when it impacts palladium in D_2O solution. It is these multiple factors -- obtained from data of several labs -- which offer strong support for the helium generation information. These reports are consistent with a nuclear origin as the explanation for some of the cold fusion phenomena.

Rothwell on CETI (continued from page 3)

The Delta T temperatures and reservoir temperatures are measured with K-Type thermocouples, with Omega Model HH22 Microprocessor Thermometers. Power is measured with Metex M 3800 series multimeters. The first test was marred by a malfunction in the control cell. The control cell consisted of tin plated steel shot, arranged as an electrochemical cathode, in the same configuration as the smaller CETI thin film beads. During tests at the lab leading up to the conference, this produced no excess heat, as expected. However, during the first test at one point it appeared to be producing a Delta T temperature as high as 2.6 deg C. Assuming the flow rate and input power were stable, this would indicate a 216 watt excess. Cravens said he thought was due to a short circuit or an obstruction in the flow, or both, since an obstruction would likely cause both problems. He turned off the control cell for safety, and repaired it later on. He told me the next day his suspicion was confirmed, the anode and cathode had come in contact because it was plugged up. I expect this explains the apparent excess, but I cannot be sure because I have no detailed data. I was not able to observe the equipment closely, verify the thermocouple readings, input power levels, flow rate, or note when the apparent excess began. (The incident occurred soon after I arrived. I was sitting across the room listening to the exposition.) The control cell was replaced with a joule heater for the remainder of the conference, which raised the water temperature the normal, expected amount.

Later on, in subsequent tests, I was able to observe the machine closely, and to make direct measurements of its performance with my own instruments. I tested the flow rate on the cold fusion cell side several times. As noted above, I did not see any measurable variation except when the flow was deliberately changed from 1,300 ml to 1,000 ml per minute by closing the valves. I checked the thermocouple readings in the reservoir, inlet and outlet with two thermistors and a thermometer. They agreed closely with the thermocouple readings. The reservoir temperature can be taken by removing the cooling loop section on top and inserting the thermistor probe directly into the water. Measuring inlet and outlet temperature required a little more ingenuity. I confirmed the outlet thermocouple reading by taking a 250 ml sample of water from the outlet pipe during a flow test and immediately measuring the temperature before the sample cooled significantly. I confirmed the cold fusion inlet temperature by turning off the control side joule heater and taking a 250 ml sample from the control outlet pipe.

CETI plans to follow up on this with demonstrations of prototype consumer products, including larger cells for space heating and heat engines. They are working to develop these devices as rapidly as they can. They estimate that it will take six months to one year to make suitable prototypes. CETI is now engaged in joint R&D projects with five corporate and university strategic partners, including the University of Illinois and the University of Missouri.

All five have independently verified the excess heat. The University of Illinois group has fabricated beads from scratch using a sputtering technique rather than electrolytic deposition. They have observed excess heat from their own beads as well as beads provided to them by CETI.

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It is the only cold fusion journal which is devoted exclusively to the material, economic, and intellectual property issues of loading isotopic fuel - a hydrogen isotope - into a material used to produce useful energy.

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THE PATENT PAGE

Analysis of the Intellectual Property

*** Returns Next Issue ***

COLD FUSION THEORIES

Given the discussions of some, it might seem that it does not look totally good for the CFT demo. These calculations by Mitchell Jones are the most interesting at this time (check sci.physics.fusion some of the latest cross-fire) and supplemental comment of Dick Blue which balance the other more numerous favorable reports. We invite CFTI to respond in quantitative fashion when their reports are out and available.

Jones' View of Anaheim: Some Simple Calculations Assuming 42 Cubic Feet per Minute by Mitchell Jones

"The problem with the Power Gen demo was not that the air-flow could not dissipate the heat. The problem was that there wasn't enough surface area of materials (mostly plastic) to deliver the needed 1344 watts to the air without a runaway temperature increase in the fluid. To see why, let's do the requisite calculations. We begin with Fourier's simple differential equation expressing heat conduction across a barrier:

$$q = k \cdot A \cdot dT/dx$$

Here q is the rate of heat transfer (e.g., in calories/sec), k is the value of thermal conductivity of the material of which the barrier is composed, A is the cross sectional area of the barrier perpendicular to the direction of heat flow, dT is the differentially small (and taken here to be positive) temperature change across a differentially small thickness dx , with dx being measured parallel to the flow of heat. To apply this to our specific situation, we begin by multiplying both sides by dx/A , giving $q \cdot (dx/A) = k \cdot dT$

Writing the definite integral, and integrating both sides, we get:

$$\int_{x_0}^{x_1} \left[\frac{q \cdot dx}{A} \right] = \int_{T_0}^{T_1} [k \cdot dT]$$

Expanding the right side and reversing the order we have:

$$[k \cdot (T_1 - T_0)] = \int_{x_0}^{x_1} \left[\frac{q \cdot dx}{A} \right]$$

Expansion of the right side depends on the specifics of the problem -- i.e., on whether A varies and how it varies. For the case of hollow cylinders with hot electrolyte on the inside and air on the outside, $A = 2 \cdot \pi \cdot x \cdot L$, where L is the length of the tubing and x is the radius.

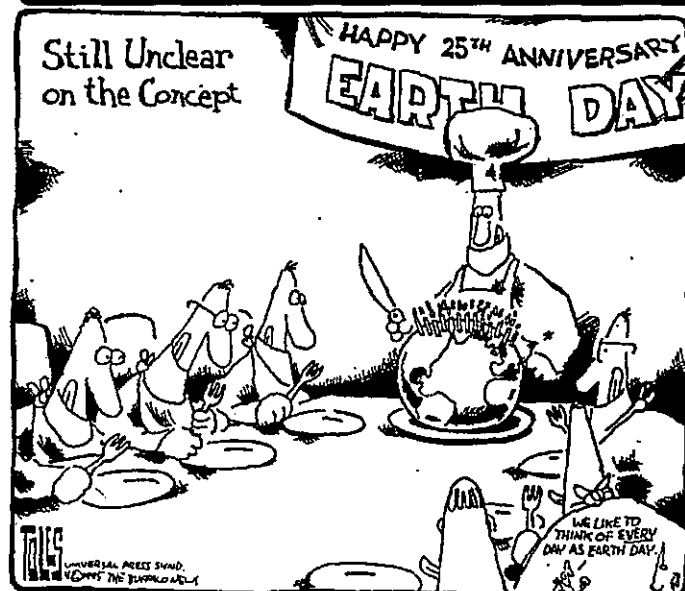
Rewriting:

$$\left[\frac{k \cdot (T_1 - T_0) \cdot (2 \cdot \pi \cdot L)}{q} \right] = \int_{x_0}^{x_1} \left[\frac{dx}{x} \right]$$

Expanding the right side, we quickly get:

$$k(T_1 - T_0)(2 \cdot \pi \cdot L)/q = \ln(x_2/x_1)$$

COLD FUSION TIMES'
FAVORITE CARTOON OF THE "MONTH"
Editorial Humor (5/16/95)



Thus, for plastic tubing, we have: $q = k(T_1 - T_0)(2 \cdot \pi \cdot L)/\ln(x_2/x_1)$

If we use A_m to represent the logarithmic mean area of the cylindrical barrier between the electrolyte and the air, this can be rewritten as:

$$q = k \cdot A_m \cdot [(T_1 - T_0)/(x_1 - x_0)]$$

In cases where the area of the outer surface of the tubing is not greater than twice as large as the area of the inner surface, we can use the arithmetic mean [to wit: $(A_1 + A_2)/2$] of the inner and outer surface areas rather than the logarithmic mean, and thereby simplify our calculation. Substituting the arithmetic mean into the formula and manipulating, we soon get:

$$q = k \cdot (A_1 + A_2) \cdot (T_1 - T_0)/[2(x_1 - x_0)]$$

Everything in the Power Gen setup consisted of hollow cylinders with hot electrolyte on the inside and air on the outside. The cell itself was a hollow glass cylinder. The tubing was essentially a hollow plastic cylinder, and so was the pump reservoir. Since we have been told that the cell outlet temperature was 36 degrees C and the inlet temperature was 20 degrees C, it is evident that the average temperature of the electrolyte in the external part of the loop (the part outside of the cell itself) was about 28 degrees C. Since the external part of the fluid loop is where the 1344 watts would have been dissipated, we can take 28 degrees C as the fluid temperature for computational purposes. Since we have assumed ambient temperature in the room to have been 20 degrees C, that becomes the temperature of the air outside the cylinders. And, assuming perfect mixing of the electrolyte inside the cylinders, it follows that the temperatures of the inside surfaces of the two cylinders (the tubing and the reservoir) would both be 28 degrees C. Similarly, assuming perfect mixing of the air on the outside of the cylinders, the temperatures of the outside surfaces would both be 20 degrees C. (Of course, perfect mixing did not occur and, thus, the real temperature gradient across these surfaces was less than 8 degrees C. As a consequence, the true heat conduction rate for the apparatus would be somewhat less than the value we are about to calculate. Our value will be the theoretical maximum conduction rate for the system, assuming idealized conditions. If that idealized maximum capacity is not sufficient to dissipate 1300 watts, you can rest assured that the actual system could not do it!) Since $T_1 - T_0 = 28 - 20 = 8$ degrees C for the external portion of the fluid loop, the formula becomes:

$$q = k \cdot (A_1 + A_2) \cdot (8)/[2(x_1 - x_0)] = 4k \cdot (A_1 + A_2)/(x_1 - x_0)$$

According to Jed we have a maximum of 18 feet of 3/8th inch ID plastic tubing in the loop. If it is thin-walled, then it is 1/2 inch OD, and the corresponding radii will be half of those values. Thus for the tubing, $x_0 = 3/16$ in = .476 cm, and $x_1 = 1/4$ in = .635 cm. Thus the area of the inside surface of the tubing is $A_1 = 2 \cdot \pi \cdot (.476) \cdot 18 \cdot 12 \cdot 2.54 = 1640$ cm², and the area of the outside surface of the tubing is $A_2 = 2 \cdot \pi \cdot (.635) \cdot 18 \cdot 12 \cdot 2.54 = 2189$ cm².

Since A_2 is less than twice the value of A_1 , our formula will estimate the conductive capacity of the plastic tubing to better than 4 percent accuracy. All we need is a conservative estimate of the thermal conductivity, k , of plastic. In my copy of *Handbook of Applied Thermal Design*, by Guyer, pg. 6-40, I find that thermal conductivities for plastics fall in the range from .05 to .2 BTU's/hr-ft-deg F.

To be very conservative, let's use the upper limit of this range for our calculations. Converting to appropriate units, we have values into the formula given above, we have 332 watts. That's a hell of a lot less than 1344 watts, now isn't it? And all we have left in the external portion of the fluid loop is the pump reservoir itself. In that case, $x_0 = 7.5$ cm and $x_1 = 7.8$ cm, and the height of the fluid in the reservoir when the pump is running is generally about 10 cm. Thus $A_1 = 2 \cdot \pi \cdot (7.5) \cdot 10 = 471$ cm², and $A_2 = 2 \cdot \pi \cdot (7.8) \cdot 10 = 490$ cm². Again, A_2 is less than twice the value of A_1 , so our formula will work just fine. Therefore, we have: $q = 4 \cdot 8.26 \cdot 10^{**} (471 + 490)/(7.8 - 7.5)$, or 44.5 watts.

The only surfaces left in the external portion of the fluid loop which could dissipate some heat would be the top and bottom of the reservoir. The top is made of thick black plastic and is separated from the water surface by an air layer that is about 10 cm thick, so its heat dissipation capacity is trivial -- not worth figuring. The bottom of the reservoir is also not worth figuring, because it rests on top of the thick plastic motor housing, separated from it by an insulating layer of trapped air. Some evaporative cooling would also be possible through a small hole that Cravens drilled in the reservoir lid to permit H₂ and O₂ gases to escape. I have tested this experimentally, and found its effect to also be trivial. (continued page 8)

ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

7

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ELEMENTARY SURVEY OF CETI MICROSPHERE DEMO

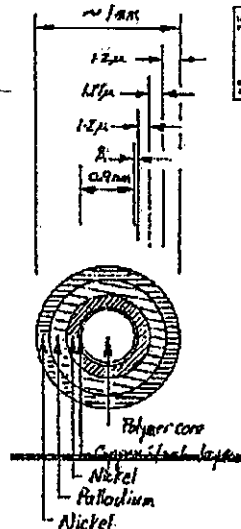
Mitchell Swartz

Why use them?

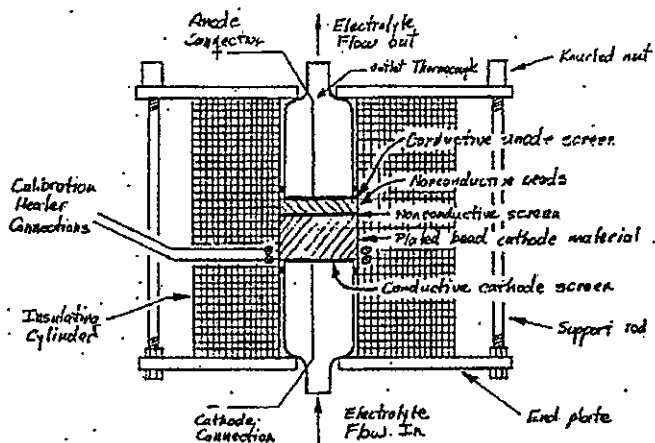
The advantage of the CETI microspheres is reported to be the use a few per cent of the metal of other systems. Another advantage the company claims is the purported ability of the beads to withstand up to 30 thousand atmospheres.

What are they?

There is limited material available except for the patent. The CETI microspheres appear to be multilayer metallic coats, used as a distributed electrode, cast over a styrene interior bead. Based upon some photomicrographs and video shown at the MIT IAP '96 meeting, the plastic surfaces appear similar to microscopic golf balls. The material purported to contain cross-linked styrene divinyl benzene. The photographic ratios appear slightly different from the drawings of Bruce Klein (courtesy of Victor Lapuszynski and John Logajan). The latter (right), and drawings of the compressive device (midcolumn, after Bruce Klein) and system assembly (top right, after Akira Kawasaki) have been laboriously assembled by them.



Ni/Pd/Ni Beads



The setup is shown on page 1, and the control box and reservoir are shown on the bottom of the page photographs taken by Akira Kawasaki and Jed Rothwell on December 6, 1995 (URLs are located on the bottom of 9). [Ed. We are assured that the next issue of *Infinite Energy* will contain further information].

The reports are that first copper is placed so as to cover over the plastic microsphere using a solution of copper chlo-

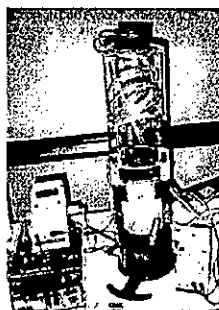
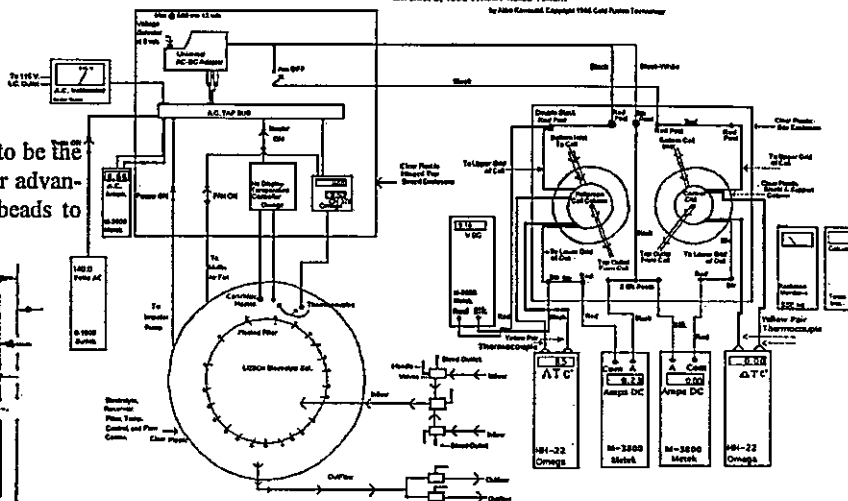


Diagram of Power-Gen '95 C.E.T.I. Patented Cell Demonstration Set-up. On Dec. 6, 1995 Whelan Philpott Talent, by Mike Gorman, Copyright 1995 Cold Fusion Technology



ride. Then a series of micron thick layers composed of nickel, then palladium, and then nickel again, are placed over the sphere using combinations of electrode-less and electrode process. The microspheres are packed together so as to create an electrode bed of size about 2 centimeters in width. The pressure is derived from pistons on two opposing sides (top and bottom of the electrolytic cell).

(continued page 9)

"ACCIDENT" IN LAB

(Continued from page 10)

In fact, he claims the motor is so efficient that, when the scooter is throttled back and free-wheeling, the engine becomes a generator and partly recharges the batteries while on the move, giving the scooter its enormous range. Michael Lughton, professor of electrical engineering at London University, [this is not the university engineer I met] is impressed. "It's an incredible machine," he says. "Takahashi seems to have developed an extraordinarily efficient electric motor and control system. In principle there's no reason why it couldn't be scaled up for an electric car." Takahashi has a good record in commercial innovation. While at Sony, he developed Beta videotape technology, which became the standard system used by the television industry until it was overtaken by VHS. He now has big plans for commercial exploitation of his new magnetic discovery.

"The YT magnet can be used for any application where conventional magnets are currently used - from credit cards to loudspeakers, with a huge potential increase in information-storage and quality," he says. One novel use for the magnet invented by Takahashi is to extend the life of the rechargeable batteries. His magnets can be made into thin inch-wide squares, which, if fitted to mobile-phone batteries, will double the amount of charge they retain and so last twice as long. This "battery doubler" is already on the market in Japan where, says Takahashi, the Japanese equivalent of BT [British Telecom] has ordered 100,000 of them.

At present the magnetic alloys are manufactured under license in Japan but last month Takahashi announced his intention to set up his primary manufacturing plant in Britain. "Britain has lower overheads than many other countries and there are hundreds of engineering companies within a few hours drive of London," he says. A factory site has already been earmarked in north London, though Takahashi now requires 20 million British Pounds [approx. \$30,000,000] investment to develop it properly.

Jones' View of Anaheim

(continued from page 6)

Conclusions

With the claimed 16 degree C temperature drop in the external portion of the fluid loop and 18 feet of plastic tubing, the Power Gen demo could have dissipated no more than about $332 + 44.5 = 376.5$ watts, which is far, far less than the 1344 watts that were calculated based on the measurements. Worse, 376.5 watts is an idealized maximum figure that assumes perfect mixing of the fluids on both sides of the barrier. It would be possible to apply film coefficients and compute the effects of imperfect mixing on these results, but that would substantially lower the calculated heat dissipation capacity of the setup. Similarly, my calculations assume that the thermal conductivities of the plastics used were at the top of the range of the known thermal conductivities for plastic. If an average value for the conductivity of plastic had been used, the potential dissipation would have been 38% less. Based on the calculations given in the previous sections, we would need at a bare minimum $(1344/376.5) \times 18 = 64.25$ feet of plastic tubing in the external portion of the flow loop to deliver 1344 watts of heat to the air with the claimed 16 degree C temperature rise!

Bottom line: The problem isn't that the air flow can't carry away the heat; the problem is that the plastic barrier material can't deliver the heat to the air fast enough to prevent a runaway temperature increase in the fluid. My theoretical analysis suggests that with 18 feet of plastic tubing, the temperature would rise far, far above the claimed equilibrium temperature of 36 degrees C, and this conclusion is confirmed by my experimental results.

P.S. I'm sorry for going into such tedious detail. I did it to encourage critical feedback. If I have made an error here, or in the experiments that I detailed in my earlier posts, I want to know about it. I would much prefer that my car ran on water and got 300,000 miles to the gallon, and I am eager for someone to prove that it is going to do so in the near future!

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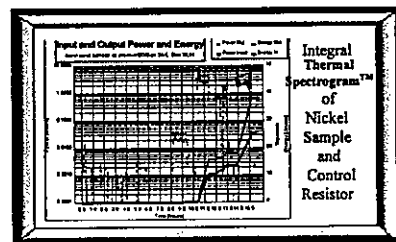
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FIRST HAND REPORT OF THE ANAHEIM CETI DEMO

by Zoltan Szakaly

[Ed.: This excerpt is from a previously enthusiastic, now disenchanted, individual.]

ANAHEIM, CA - I went to see the CETI cold fusion demo at the Anaheim Power-Gen conference. .. I drove to Anaheim which took me over an hour due to the traffic jams and accidents and all. This was on Tuesday, in the morning, I was at the site about 10 am. .. When I got to the CETI booth I found Jed Rothwell and several people from CETI, including Dennis Cravens. They told me very enthusiastically that the reactor had been producing 1400 watts of power on Monday in their hotel room. I looked at the experimental setup and saw a large pump, about 5 inches diameter, in a vertical glass tube. A filter was visible besides the pump in this container. The type of pump it was could not be made out, which was a disappointment. Apparently the pump was submerged in the electrolyte. It may have been a peristaltic type of pump, I don't know. There was an instrument which looked like a voltmeter but it had two thermocouple inputs. This showed the supposed temperature difference. There were two vertical columns, one with the famous beads, the other was the control. There were four meters hooked up to show voltage and current of the cells.

The CETI people told me that the cells were turned off, only the pump was running. The temperature gage was constantly blinking, Jed said this was because it was on some funny scanning mode. I saw temperature readings from about 0.5 to 4.5, constantly changing. They told me that they could not turn on the cells because the control cell was clogged up. They swore that the previous day the cell was producing 1400 watts. They would not say how long. They said it was so hot that they had to turn on the air conditioning in the room. When I asked them to turn on the cell they said they might turn it on around three in the afternoon. My general impression of the experimental setup was poor. The circulating liquid had large bubbles of gas in it, about two-three inch long sections of the half inch diameter transparent tubing. I was highly unimpressed by the thermocouples. There was no way of telling why the reading kept fluctuating wildly or what the reading meant, like centigrade or Fahrenheit or what. I could not tell what type of pump they had or what they had inside their electrolytic cells. The cells were covered by non-transparent insulation. I would have preferred to see a clearly visible peristaltic pump, transparent cells and clear temperature readings. There was no way to tell what the flow rate was. They had a glass bottle, so they could pull off the tubing and let the liquid run into the bottle for a minute, to see the flow rate. The bottle had non-parallel sides, it was one of those glass things that taper towards the top. I did not feel that I could trust that to measure the flow rate. .. I could not observe any heat production, but of course the cells were not on.

When I asked questions I was met with distrust, the CETI people did not seem to take to me. I left totally frustrated.

Later on my way home and at home I decided that perhaps I had better give these people a second chance, so I returned Wednesday morning. I got to the site around 10:15 am. This second time there was one guy from CETI there. The equipment was missing. This guy was very friendly, and he told me that they were producing about 400 watts Tuesday afternoon. He told me that they took the entire setup back to the Hilton so that they could give private demos. I think it is highly unprofessional of them to rent a booth at a conference and then not display the equipment. If the equipment does not work they should not rent a booth. .. We could say that this is difficult technology and it does not always work right. I think claims that the CETI cells have been producing heat for weeks are exaggerated. My earlier impression was that the CETI cell had been running at various conferences

OP-ED COLUMN -- WHAT'S HAPPENIN' Winter '96 and NEWS from the INTERNET (also page 3 and 6)

and had been left on for weeks. Now I got the impression that they could only run the experiment when I was not there to see it. They could only run for short periods of time, in which case the heat could simply be explained by the burning of hydrogen accumulated in the electrolyte. I certainly felt glad that I was not holding CETI stock. ➤

ELEMENTARY REVIEW (continued from page 7)

In the electrolytic cell, as a result of the opposing surfaces, the beads are crushed together to form one of the two required electrodes and water percolates through, thus removing the heat. The stream of hot water exits from the top. A platinum screen is used for the contacts to the microspherical array which is a distributed cathode. CETI reports to use 1 to 2 molar lithium sulfate. Our close examination of photomicrographs of the beads after their use reveals significant size heterogeneity, and what appears to be observations of some green areas on the surfaces, other surfaces were cracked, with flaked-off layers. The green may be consistent with a chemical form of oxidized copper (perhaps such as patina).

There have been several complaints about the CETI demonstrations regarding recombination (to be covered in the next issue) and the calorimetry (partially reviewed on pages 3 and 6). However, we believe that although flow measurement made with changes in the pressure head may not reflect actual higher resistance under other laboratory or test conditions, and the role of other fluid mechanical (Bernard) instabilities may have not been mentioned or considered, the demonstration does appear as probable progress in the rapidly developing cold fusion field. ➤

The COLD FUSION Electric Car

by Dana R. Rotegard MCFA

Major work on Cold Fusion as a power source for electric cars is underway in the R&D labs of several major international auto manufacturers led by Toyota, the sponsors of Dr. Fleischmann and Pons lab near Nice, France. The American "Big Three" of GM Ford and Chrysler continue to fight electric car mandates in Massachusetts, New York, and California. Quite clearly the executives of American auto companies are ignoring the example of their overseas competition and cutting edge scientific developments that could make internal combustion technology economically obsolete.

This summer, a coalition of entrepreneurs, economic futurists, environmental groups such as MASSPIRG, and medical lobbies such as the American Lung Association successfully defended the electric car mandates from an attempted rollback by Detroit. Cold fusion and breakthroughs in zero point energy technology make it possible that market forces will soon end the age of fossil fuel.

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"ACCIDENT" IN LAB CREATES SUPER MOTOR

As reported Sunday Times (10 December) by Tony Ewards

London, UK, Japan -- An electric scooter with a top speed of 50 mph and a range of more than 500 miles has been developed by a Japanese scientist who accidentally discovered what he claims is the world's most magnetic material. To date, most of the research on electric vehicles has concentrated on developing super-efficient batteries in an attempt to maximize their range and power-to-weight ratio. However, until now even the most advanced vehicles have required a small battalion of such batteries to achieve a modest

performance. The new scooter, developed by Sciex Corporation of Japan, runs on just four ordinary 12-volt car batteries. "Almost everyone has worked on the battery end of the problem," says its inventor, Yasunori Takahashi. "I thought: why not look at the other end - the motor?" His breakthrough in electromagnetic technology came a few years ago while he was experimenting with new magnetic alloys. One of his laboratory staff misread his instructions and added the wrong element to the mix. We Japanese often confuse the Roman letters b and d," says Takahashi, "My technician added neodymium (Nd) instead of niobium (Nb). The result was extraordinary - I suddenly found myself in the presence of the most powerful magnetic material I had ever seen." Takahashi subsequently developed a manufacturing system for producing a magnetic powder which could be formed into anything, from ultra-thin coatings to large permanent magnets. He now claims to have produced a magnet with the world's highest Megagauss Oersted rating - or MgOe, the unit in which magnetism is measured. "Before my discovery, 55 MgOe was the maximum anybody had achieved; but my magnet can reach 120 MgOe," says Takahashi. This super-magnetic force is the secret behind the new Sciex scooter's performance. Takahashi has redesigned a conventional electric motor and fitted his super-powerful "YT" magnets, resulting in highly efficient engine that will produce a claimed 15 horsepower [11 kW] from just a few amperes of electricity. (Continued on page 7)

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